

Abstract

This study compares GPT-4o's performance in rating image descriptions against human ratings using two established datasets: VICR dataset and Flickr8k dataset. By applying Kendall's Tau correlation method, the alignment between GPT-4o's ratings and human ratings is measured to assess rating quality. Based on the performance of GPT-4o, it may serve as a scalable alternative to human evaluators, potentially reducing the time and resources required for large-scale caption rating.

Background/ Methodology / Approach

- **Data Preparation:** Image-caption pairs from the VICR and Flickr8k datasets are fed into OpenAI's batch API using GPT-4o. Each image description is rated based on scales provided by VICR and Flickr8k.
- **Prompt Engineering:** Specific prompts are crafted to instruct GPT-4o to rate captions according to dataset-defined criteria, ensuring consistency across evaluations.
- **Statistical Analysis:** Kendall's Tau correlation coefficient, statistical measurement of the ordinal association between two measured quantities, is used to compare GPT-4o's ratings against human ratings, providing a quantitative measure of alignment between AI-generated and human-generated ratings.

VICR Scale

1. Objects are incorrectly identified. The caption gives the wrong idea about what is happening in the image.
2. Objects are partially correctly identified with some errors, but the caption is accurate enough to give an idea of what is happening in the image. The caption identifies most of the objects but might not identify everything. There is no interpretation of what anything means.
3. Relevant objects are correctly identified. The caption describes what is seen but not where objects are in space. There is no description of the overall setting and no interpretation of an event.
4. Objects and/or a general scene and/or an action are correctly identified but not every element is completely identified. The caption describes what is seen and where things are in space. There is no interpretation of an event.
5. Objects, a general scene, and actions are correctly identified if present in the image. The caption describes what is seen and where things are in space. Interpretation of overall setting and/or event is included.

Flickr8k Scale

1. Is unrelated to the image.
2. Is somewhat related to the image.
3. Describes the image with minor errors.
4. Describes the image without any errors.

Figure 1: Rating Scales for VICR and Flickr8k

Introduction

- **Problem Statement:** Human-evaluated captions are challenging due to time, subjectivity, and limited expert raters.
- **Research Question:** Can AI replace human raters to enhance image captioning quality and efficiency?
- **Hypothesis:** GPT-4o's ratings will align closely with human ratings, showing high correlation via Kendall's Tau.
- **Motivation:** AI raters could boost the efficiency, consistency, and scalability of caption evaluation.

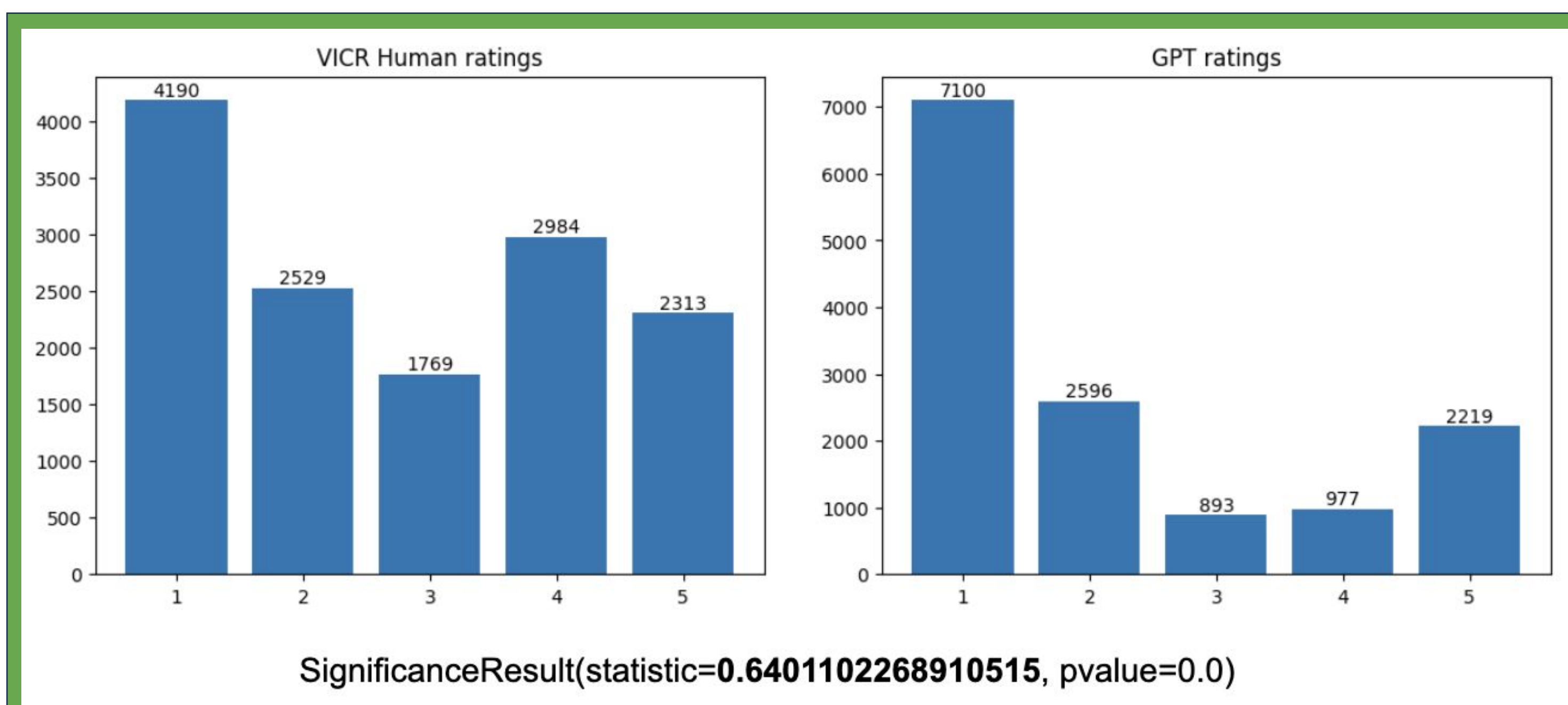


Figure 2: Histogram of VICR dataset: human ratings vs gpt-4o ratings

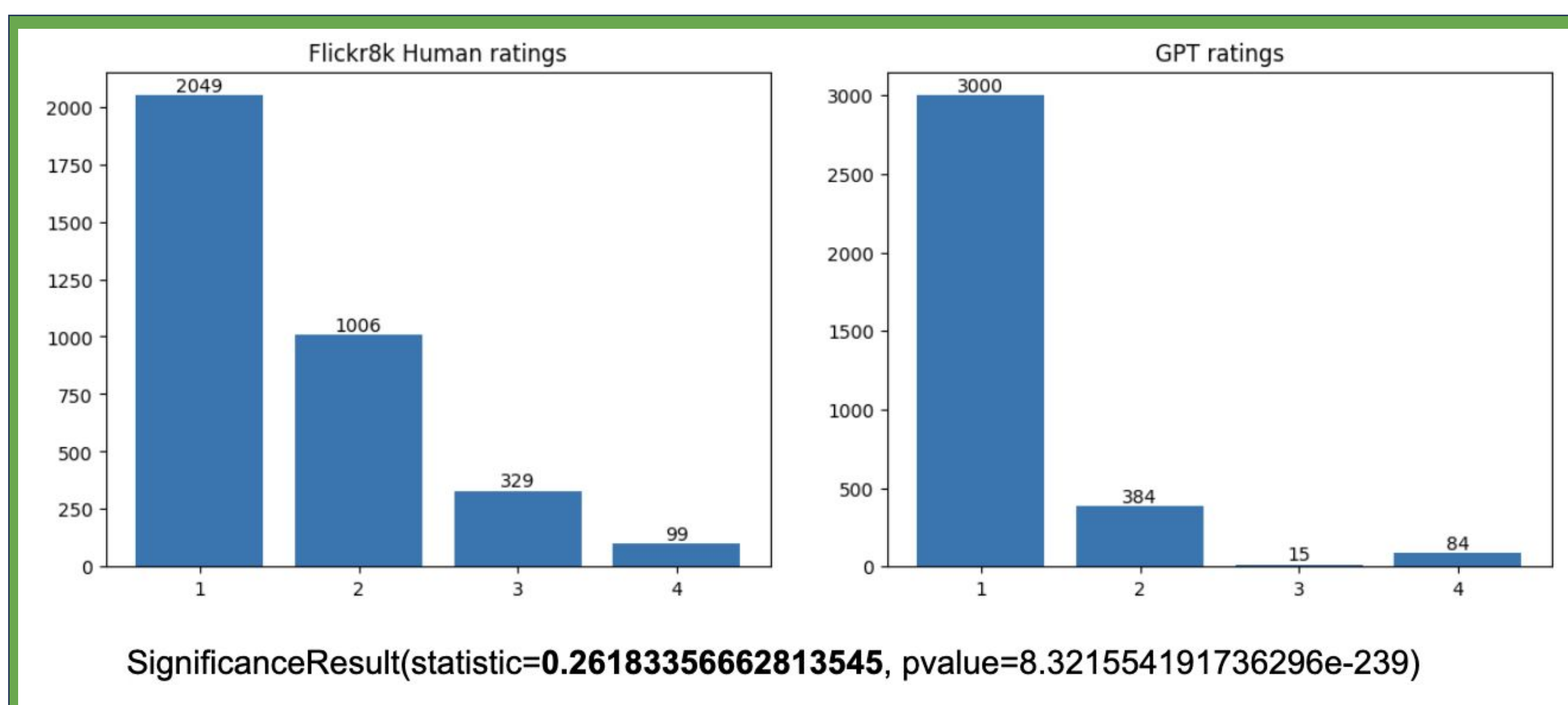


Figure 3: Histogram of Flickr8k dataset: human ratings vs gpt-4o ratings

Future Work

- **Prompt Refinements:** Explore various prompt inputs to optimize output ratings and enhance caption evaluation quality.
- **Comparative AI Ratings:** Evaluate different AI models to identify those with the potential to outperform current image caption raters.
- **Video Caption Evaluation:** Extend the study to video caption ratings, aiming to integrate findings with the YouDescribe Dataset to enhance video description generation and augmentation.

Table 4: Kendall's τ correlation with ground-truth ratings on a test subset of each dataset for various metrics and predictors.

Reference-based	Flickr8k	VICR
BLEU-1	33.7	55.8
BLEU-4	31.6	51.8
METEOR	40.4	60.2
ROUGE	33.4	53.2
CIDEr	44.1	66.6
SPICE	41.2	60.4
RefCLIPScore	51.9	71.7
ViLBERTScore	50.1	66.9
Reference-free		
CLIPScore	50.7	67.3
VSEPP	48.6	65.1
VBAIignment	49.9	65.8
Model _{Flickr8k}	53.7	71.8
Model _{VICR}	53.1	75.8



Caption: A bunch of vegetables on a countertop.
Human Ratings: 5, 5, 5, 5, 5

GPT-4o Rating: 4

Reasoning: The caption gives a general idea but misses the opportunity to include more specifics about the variety of items present. Still, it conveys the main scene correctly.

Figure 4: Kendall's Tau metrics and Qualitative result

Results / Discussion

- **High Kendall's Tau Score:** When comparing GPT-4o ratings with VICR human ratings, the results show a score of 64.0, positioning GPT-4o's performance between that of the SPICE and VSEPP metrics. For the Flickr8k dataset, however, the score was lower at 26.2, likely due to an insufficiently detailed rating scale and uneven data distribution within the dataset.
- **Bias Towards Lower Ratings:** It was observed that GPT-4o tends to assign lower ratings to image-caption pairs, particularly in aspects like object detection, scene description, and fine details, reflecting a bias towards stricter accuracy in image description.

"a close up of a person holding a carrot"

Human ratings : 4, 5, 5, 4, 4

GPT rating : 2 → 5

The description identifies a person holding a carrot, but it inaccurately suggests there's only one carrot when multiple are being held. It also misses key details like the dirt on the carrots and their freshly harvested state.



Figure 5: Qualitative result with lenient prompt

Summary / Conclusion

This study demonstrates that GPT-4o can serve as an effective tool for rating image descriptions, potentially reducing reliance on human evaluators while maintaining rating quality. The findings show that GPT-4o, AI-based evaluators, can mirror human judgment in image captioning tasks, provided that the rating scale is sufficiently well-defined.

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